NucMed Board Studio · protocol study sheet

SPARROW

RM2-3301 · ^{68}Ga-LNTH-2401 GRPR PET · Phase 2/3

Lantheus · IND 177584 · 100 pages

CLINICAL STUDY PROTOCOL

Protocol Title:	A Prospective Phase 2/3, Multicenter, Open Label Study of [^{68}Ga]-LNTH2401 PET/CT Imaging in Participants with Prostate Cancer: Phase 2 Dose Optimization in either Newly Diagnosed or Biochemically Recurrent Prostate Cancer and Phase 3 Diagnostic Performance in Participants who are Newly Diagnosed and Have at least High Risk or Suspected Metastatic Disease and Planning to Undergo Radical Prostatectomy Plus Pelvic Lymph Node Dissection (SPARROW)
Protocol Number	RM2-3301
Investigational Product:	[^{68}Ga]-LNTH-2401
Indication:	Prostate Cancer
Study Phase:	Phase 2/3
Brief Title:	Phase 2/3 [^{68}Ga]-LNTH-2401 PET/CT Dose Optimization and Diagnostic Performance in Prostate Cancer (SPARROW [Staging Prostate Cancer with Advanced RM2 Radiotracer Oncologic Workup]).
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Study Identifier Number(s)	US – IND Number 177584 (FDA)
Protocol Version Number and Date	Version 1.0, 30 April 2026

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What this study is, in one breath

Inject a GRPR (bombesin) PET tracer, pick the dose in Phase 2, then in Phase 3 ask whether pelvic-node pictures match prostatectomy pathology.

Diagnostic PET. Not PSMA. Not a therapy.

> **If the protocol feels like a wall of words**
>
> A protocol is a recipe: who can join, what is injected, how much, when you scan or treat, and how you decide if a picture or outcome was right. This sheet is the same idea in normal English. It does not replace the signed site copy.

1. Not a PSMA agent

Lantheus SPARROW / RM2-3301 is ^{68}Ga-LNTH-2401 (DOTA-RM2), a GRPR / bombesin-family PET tracer for prostate cancer. Complementary to PSMA, not a substitute. IND 177584. Precursor ≤ 40 µg. Sterile ready-to-use, pH 4.0–6.5.

- Phase 2 start: 5 ± 1 mCi (148–222 MBq). Labelled range 3–6 mCi (111–222 MBq).
- Slow IV bolus + saline flush. Single dose. No planned dose modifications.
- Do not call this PSMA-11, Pylarify, or Pluvicto.

2. Two sequential parts, no randomisation

Part	Who / n	What it answers
Phase 2 dose opt	≤12 imaging-evaluable GRPR+ (ND or BCR)	Pick activity + time window. Serial PET 15/30/60/120 min. IQS ≥ 10.5.
Phase 3 pivotal	~396 at ≤30 US sites; total study ~408	High-risk newly diagnosed, planning RP+PLND. Histopathology is SOT.

Formulation 1 in Phase 2, Formulation 2 (commercial-intent) in Phase 3. iDMC reviews the chosen activity/window before Phase 3. Phase 2 dosimetry does not feed the Phase 3 primary analysis.

3. Phase 3 primary

Participant-level pelvic lymph-node sensitivity and specificity vs RP+PLND pathology. Three blinded central readers. 95% CI lower bound targets: sensitivity ≥ 25%, specificity ≥ 65%. Primary analysis set = everyone who got the tracer (intent-to-image).

Secondary composite SOT can add conventional imaging and PSMA PET under an iDMC charter. Newly diagnosed, at least high-risk or suspected M1, planning RP+PLND.

4. NM handling

- Hydrate before and after. No high-energy (>300 keV) gamma tracer within 5 physical half-lives.
- No IV iodinated contrast within 24 h; no high-density oral contrast within 5 days (water contrast OK).
- Prior anti-cancer therapy or investigational agent within 28 days / 5 half-lives = ineligible. EBRT only in Phase 2 BCR.

What to know cold

Q. Target?
A. GRPR / bombesin (DOTA-RM2). Complementary to PSMA.

Q. Phase 2 dose?
A. Start 5 ± 1 mCi (148–222 MBq). Lowest activity with mean IQS ≥ 10.5.

Q. Phase 3 n?
A. ~396 high-risk ND heading to RP+PLND. Total study ~408.

Q. Primary?
A. Participant-level pelvic LN sensitivity (CI ≥ 25%) and specificity (CI ≥ 65%) vs pathology.

Q. Readers?
A. Three blinded central readers. Path is the SOT.

How to use this with Board Studio

Open the matching protocol module and press play. The left pane is the signed page. Use this PDF when legalese stalls you. If anything conflicts, the signed protocol wins.

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